

FIT PASSPORT

Detailed Start-Up Proposal

49-800 Start Up Creation in Practice | Fall 2026

One body. One fit identity. Any store.

A consumer-owned fit layer for apparel, with a path to footwear, jewelry, and pet sizing.

Semester MVP: responsive web app focused on human apparel. **Exploratory extension:** dog/cat fit profiles.
Future roadmap: footwear, rings/jewelry, browser extension, and visual try-on.

Prepared as a detailed supplement to the official Start-Up Creation in Practice proposal form.

1. Executive Summary

Online shoppers still have to relearn their size every time they move to a new brand, retailer, country, or product category. A size label such as M, 32, EU 48, or US 10 is only meaningful inside a specific sizing system, and even within a brand the cut and fit can change by style. Research has long identified fit and size uncertainty as a meaningful source of concern in online apparel shopping, and fit information can affect return behavior and purchase confidence.[2][3][4]

Fit Passport is a consumer-owned fit identity. Instead of rebuilding a profile at every store, the user keeps one portable record of body information, preferred fit, products that already fit well, and real outcomes from purchases, returns, and exchanges. When the user finds a product on another retailer site, Fit Passport reads the product information, normalizes the relevant sizing data, and recommends a size with an explanation and confidence level.

The semester project will deliberately stay narrow. The core MVP is a responsive web application for human apparel, initially tops, shirts, and jackets. A user can create a profile, add several known-good garments, paste a product URL, receive a recommendation, and later report whether the product fit. A small pet profile prototype will test whether the same architecture can support dog or cat apparel and harness sizing. Footwear, rings, jewelry, a browser extension, and visual try-on are part of the product roadmap rather than required MVP deliverables.

The startup thesis is not that fit technology does not exist. It clearly does. True Fit, Bold Metrics, Sizebay, 3DLOOK, Google, and Perfect Corp demonstrate strong demand and technical feasibility across size recommendation, body measurement, virtual try-on, and digital sizing.[6][7][8][9][10][11] The opportunity we want to test is different: whether fit intelligence can become a user-owned layer that travels with the shopper, rather than remaining primarily embedded inside individual retailer or vendor ecosystems.

2. The Problem We Want to Solve

The consumer problem sounds simple: “What size should I buy?” In practice it is a repeated decision problem with incomplete information. Shoppers combine brand size charts, model measurements, product reviews, remembered experience from other brands, garment material, preferred looseness, and sometimes multiple orders of the same item. The underlying issue is not only body size; it is the relationship between a particular body, a particular garment, and the way that person wants the garment to fit.

Research on apparel e-commerce supports this framing. Studies have identified concerns around imagining fit, inability to try items on, and uncertainty about size as distinct online shopping concerns.[2] Experimental work on virtual size selection has also treated fit as a central problem because shoppers must choose without physically trying the garment.[3] Research on online returns further shows that contextual fit information can help shoppers interpret whether a product is appropriate for them and can reduce returns.[4]

The business cost is also material. The National Retail Federation projected \$849.9 billion in U.S. retail returns in 2025 and estimated that 19.3% of online sales would be returned.[1] Not all of those returns are caused by sizing, and this proposal does not assume they are. The point is that return friction is already large enough that better purchase confidence has clear economic value.

2.1 What is fragmented today?

- Identity is fragmented: shoppers maintain separate retailer accounts, saved sizes, notes, receipts, and memories.
- Sizing language is fragmented: US, EU, UK, numeric, alpha, shoe, and ring systems do not map cleanly across brands.
- Fit preferences are personal: two people with the same body measurements may want very different silhouettes.
- Outcome data is wasted: a return because “shoulders were tight” is valuable personal fit data, but the shopper rarely owns it in reusable form.
- Product data is inconsistent: some pages publish garment measurements, some body charts, some only model height and size, and some only qualitative fit notes.

3. Product Concept: A Portable Fit Identity

Fit Passport treats fit as an identity that improves over time. The first version does not try to create a perfect 3D body model. It starts with low-friction information users actually know: height, a few optional measurements, fit preference, and several pieces of clothing that already fit well. The system then combines that personal context with product-specific sizing information.

3.1 Core user flow

Step	What happens
1. Create a Fit Passport	Enter basic body data, preferred fit (slim / regular / relaxed / oversized), and optional measurements.
2. Add known-good items	Add 3-5 garments that already fit well: brand, category, size, and a short fit note.
3. Paste a product URL	The system extracts brand, category, available sizes, size chart, model sizing, fit notes, material/stretch, and regional sizing when available.
4. Get a recommendation	Fit Passport ranks the available sizes, returns a recommended size, confidence, and a short explanation by body area or comparable known-good items.
5. Close the loop	After purchase, the user records keep / return / exchange and specific fit feedback. That outcome becomes personal ground truth for later recommendations.

3.2 Example recommendation

Recommended size: EU 48 Confidence: 86% Shoulders: comfortable Chest: slightly relaxed Sleeve: about 1-2 cm longer than your usual preference Why: closest match to your COS M and Uniqlo L history; this product is cut smaller through the chest than those reference items.

3.3 The pet extension

The same architecture can be tested on pets without turning the project into a second full product. A dog or cat profile can store species, breed/body shape, weight, neck, chest/girth, back length, and known-good harnesses or apparel. Mainstream pet brands still ask owners to measure body dimensions manually; Ruffwear, for example, uses chest girth as a key sizing input for harnesses, apparel, and packs.[14]

The value of the pet experiment is architectural: can the same model of body identity + product attributes + known-good examples + fit outcomes work in another sizing domain? The semester deliverable is only a small prototype and 3-5 pet-owner interviews, not a production pet-sizing engine.

4. Why Now and Market Opportunity

Fit technology is moving quickly because several enabling capabilities have matured at the same time: AI-based body modeling, structured product extraction, generative virtual try-on, and retailer demand to reduce uncertainty and returns. The global virtual fitting room market is estimated at about \$8.5 billion in 2026 and is forecast to reach \$20.6 billion by 2030.[5] This is not the exact revenue market for Fit Passport, but it is strong evidence that retailers and consumers are spending around the broader fit/try-on problem.

The pet extension also sits in a real adjacent market. Grand View Research estimates the global pet clothing market at about \$2.3 billion in 2026.[13] APPA reports that 95 million U.S. households owned at least one pet in

2025; 71 million households owned dogs and 53 million owned cats.[12] The extension is therefore large enough to test, while still remaining secondary to the apparel beachhead.

Signal	Evidence	Why it matters
Online returns	\$849.9B projected U.S. retail returns in 2025; 19.3% of online sales returned.[1]	Purchase confidence and fit mistakes have meaningful economic cost.
Fit-tech category	\$8.5B global virtual fitting room market estimate for 2026; \$20.6B forecast for 2030. [5]	Existing demand validates the problem space.
Pet adjacency	\$2.3B global pet clothing market estimate for 2026.[13]	Pet fit can be an adjacent vertical, not just a novelty feature.
Pet ownership	95M U.S. pet-owning households; 71M dog and 53M cat households in 2025.[12]	Large reachable consumer base for future expansion.

5. Target Customers and Beachhead

5.1 Primary persona: cross-brand online shopper

A 18-40 year-old online shopper who regularly buys from multiple brands or international retailers. They know some reference sizes (“I wear L in Uniqlo” or “EU 48 in this brand”) but still hesitate on unfamiliar products. They may read reviews, compare model measurements, order two sizes, or avoid a purchase because they do not trust the fit.

The beachhead is intentionally narrower than “everyone who buys clothes.” We want users with repeated fit decisions and enough prior purchase experience to provide known-good reference items.

5.2 Secondary persona: pet owner

A dog or cat owner who buys harnesses, coats, sweaters, recovery clothing, collars, or other size-dependent gear. They may know that one brand’s size S fits, but still need to remeasure for a new brand or product. The pet user is useful for testing cross-domain portability, not for expanding the MVP scope.

6. Competitive Landscape

The competitive landscape is strong, which is a positive signal for demand but forces us to be specific about what Fit Passport is testing. We are not claiming to have better body scanning than 3DLOOK or Bold Metrics, a larger dataset than True Fit, or a more realistic virtual try-on than Google. The wedge is ownership and portability of the fit profile.

Competitor	What it proves / does well	Opening Fit Passport explores
True Fit	Large fit intelligence network built from purchase/return outcomes; personalized size and fit guidance across a large retailer/brand dataset.[6]	Primarily a commerce/retailer infrastructure model. Fit Passport tests a user-owned profile that can be applied to a product even when a retailer has not integrated our service.
Bold Metrics	Digital twins; 50+ body data points from 4-6 simple inputs; 250M+ digital twins and 750M+ fit simulations.[7]	Very strong body measurement infrastructure. We can use low-friction inputs and focus on personal fit history rather than competing on body-model scale.
3DLOOK	Two smartphone photos can generate body measurements and structured body data.[8]	Validates phone-based measurement as a future option. Photo scanning can be an integration or later capability, not our semester moat.
Sizebay	Retailer-integrated size recommendations, regional size charts, footwear support, and fit	Strong merchant-side sizing tool. Fit Passport focuses on a reusable consumer

	preference guidance.[9]	profile and any-store workflow.
Google Doppl / Shopping VTO	Digital self, outfit visualization, and AI-generated video. Google explicitly notes fit and appearance may not always be accurate.[10]	Excellent visual “look” layer, but visual realism is different from deciding which physical size to buy.
Perfect Corp	Web-based ring sizing and AR ring try-on with regional size conversion.[11]	Validates jewelry as an adjacent sizing vertical; no need for Fit Passport to rebuild this in the MVP.

7. Core Differentiation

- 1. Consumer-owned identity:** The profile is centered on the shopper rather than a particular retailer. The long-term concept is that the same fit history travels with the user.
- 2. Any-store workflow:** The MVP starts with a pasted product URL. A retailer does not need to install Fit Passport before a user can test the product concept.
- 3. Fit experience, not only measurements:** The system learns what “good fit” means for this person from known-good garments and detailed outcomes such as chest tight, sleeves long, or too oversized.
- 4. Outcome-based learning:** Keep, return, and exchange decisions become personal ground truth. The product should improve as the user shops.
- 5. Explainability:** The recommendation should say why it chose a size and what tradeoffs to expect rather than returning an unexplained label.
- 6. Cross-domain fit architecture:** The long-term platform thesis is one household of fit identities that can extend to shoes, rings, pet gear, and other size-dependent categories.

8. Research Questions, Hypotheses, and Methodology

The course project should produce evidence, not just a demo. The following hypotheses are deliberately testable within one semester.

Hypothesis	How we will test it	Primary evidence
H1: Cross-brand sizing creates meaningful friction.	20-25 semi-structured shopper interviews + short survey.	Frequency of uncertainty, bracketing, abandoned purchases, returns, workarounds.
H2: A portable profile is more useful than a generic chart.	Concept/Figma tests comparing retailer chart-only vs Fit Passport workflow.	Purchase confidence, preference, perceived usefulness.
H3: Known-good items + preferences improve recommendations.	Blind/back-test products users already own; compare simple baseline vs personalized model.	Agreement with actual owned size / self-reported fit outcome.
H4: Outcome feedback improves later recommendations.	Re-score or re-run recommendations after several labeled fit outcomes.	Change in recommendation agreement/confidence and user trust.
H5: The architecture can generalize to pets.	3-5 pet-owner interviews + small dog/cat profile prototype.	Whether the same profile/product/outcome model remains understandable and useful.

8.1 Research sequence

- Exploratory interviews before building: understand how people currently decide size, what data they trust, and why they return items.
- Concept validation: show low-fidelity flows before implementation so feature choices are grounded in user behavior.
- Usability validation: test Figma with 8-10 people before building the full UI.
- MVP beta: 10-15 users with known products and a limited set of supported retailer pages.

- Back-testing: ask users to provide items whose real fit they already know, hide the outcome from the model, and compare recommendations against a simple baseline.
- Pet experiment: treat it as a separate mini-study, not an additional production promise.

9. MVP Scope and Product Design

The MVP should be a responsive web application, not a native mobile app. A web product matches the core action - pasting or sharing a product URL - and reduces development overhead. It is also easier to demo in class and creates a clean path to a browser extension later. Mobile photo capture and native integrations can be explored after the core fit loop is validated.

In MVP	Stretch if time	Out of semester scope
Human apparel: tops, shirts, jackets	Pet profile prototype	Full pet recommendation engine
Basic body profile + preferred fit	Regional sizing normalization	All apparel categories
Known-good garment history	Product comparison	Footwear production support
Paste product URL	Simple VTO API proof of concept	Ring/jewelry production support
Product/size-chart extraction	Photo-based body measurement	Custom body-scanning model
Size recommendation + explanation	Browser extension proof of concept	Native iOS + Android apps
Keep/return/exchange feedback	Household multi-profile concept	Retailer enterprise integrations

9.1 Proposed screens

- Onboarding: height, weight, optional measurements, preferred fit, and privacy controls.
- Known-good closet: add reference garments with brand, category, size, and fit notes.
- Check a product: paste URL; review extracted product/size data before recommendation.
- Recommendation: recommended size, confidence, expected fit by area, and explanation tied to known-good references.
- Fit history: kept / returned / exchanged products and reasons.
- Pet profile prototype: species, weight, neck, girth, back length, body notes, and known-good gear.

10. Technical Approach

The architecture should separate product extraction from fit recommendation. LLMs are useful for turning messy product-page text into structured attributes, but the recommendation itself should be as transparent and testable as possible. This avoids a black-box workflow in which an LLM simply guesses a size.

Layer	Semester approach
Frontend	Responsive React / Next.js web app.
Backend	Lightweight API layer (Next.js API or FastAPI; final choice can follow team familiarity).
Database	PostgreSQL or equivalent relational store for profiles, products, size options, recommendations, and outcomes.
Product parser	LLM/VLM-assisted extraction from public product information into a normalized schema; user can correct extraction errors.
Fit engine	Rule/score-based baseline combining available body measurements, size-chart compatibility, known-good anchors, preferred fit, and labeled outcomes.
Explanation layer	Template or LLM-generated natural-language explanation

	grounded in the fit-engine output.
Analytics	Event tracking for onboarding completion, recommendations, corrections, confidence ratings, and outcome feedback.
Optional VTO	External API or existing service only; no custom image model training required.

10.1 Core data model

Entity	Example fields
FitProfile	user_id, height, weight, optional body measurements, preferred_fit
KnownGoodItem	brand, category, size, region, fit_rating, area-specific notes
Product	retailer, brand, URL, category, material, fit notes, model information
SizeOption	size label, region, garment/body measurements, normalized attributes
FitRecommendation	ranked size, confidence, reasons, assumptions
FitOutcome	keep/return/exchange, actual size, fit issues, notes
PetProfile	species, breed/body type, weight, neck, girth, back length, known-good gear

10.2 Recommendation logic for the MVP

The first model does not need to be a proprietary machine-learning system. A transparent scoring model is preferable for the course because it can be evaluated. Each available size can receive a score based on:

- Compatibility with any available body or garment measurements.
- Similarity to known-good items in the same category or a comparable brand/style.
- Fit preference (slim, regular, relaxed, oversized).
- Regional/brand normalization signals from the product page.
- Historical penalties or bonuses learned from labeled outcomes (for example, repeated “sleeves too long” feedback).
- Data completeness; confidence should decrease when the retailer provides little sizing information.

A useful outcome of the semester may be learning that some of these signals add little value. That is still a valid startup finding because it clarifies what data users actually need to maintain.

11. Size Recommendation vs. Virtual Try-On

Fit Passport should separate “Which size should I buy?” from “What will this look like on me?” Google Doppl demonstrates that virtual try-on can produce an animated digital version of a user wearing an outfit, but Google also warns that fit, appearance, and clothing details may not always be accurate.[10] Recent research likewise treats physical fit fidelity as a distinct problem from visual realism.[15][16]

For that reason, virtual try-on is a visualization layer, not the evidence used for the size recommendation. If time permits, the project can call an existing VTO service to show how the two layers could work together: Fit Engine recommends size; Look Engine visualizes style. The semester does not require training a custom diffusion or 3D model.

12. Pet Fit Passport Extension

The pet extension makes the platform thesis more interesting without requiring a large implementation. It tests whether “fit identity” is a reusable product abstraction rather than an apparel-only feature. The profile can change its measurement vocabulary by species while preserving the same data pattern: body attributes, product dimensions, known-good items, and outcome feedback.

Human apparel	Dog/cat gear
Height, chest, waist, shoulder, inseam	Weight, neck, chest/girth, back length
Slim / regular / relaxed / oversized preference	Snug / regular / loose; body-shape notes
Known-good shirt/jacket sizes	Known-good harness/coat sizes
Return: shoulders tight, sleeves long	Return: chest tight, back short, neck loose
Product size chart + model data	Pet size chart + breed/body notes

APPA’s current ownership data and the size of the pet clothing category make this a credible future segment.[12] [13] However, the semester goal is only to learn whether owners see value in a persistent profile and whether the same data model remains intuitive.

13. Business Model Hypothesis

The business model is intentionally a hypothesis to be tested during customer discovery. The product should first prove that users will maintain a fit identity and trust its recommendations.

Canvas element	Working hypothesis
Customer segments	Frequent online apparel shoppers first; later households, footwear/jewelry shoppers, and pet owners.
Value proposition	One reusable fit identity that learns what fits you and travels across stores.
Channels	Responsive web; later browser extension, mobile app, affiliate links, and commerce APIs.
Customer relationship	Self-service profile that improves through purchase feedback.
Revenue streams	Freemium subscription; premium multi-profile/advanced features; affiliate revenue; later B2B/API licensing.
Key activities	Product-data extraction, recommendation, customer research, fit-history learning, and UX.
Key resources	Fit profiles, structured product data, user-labeled outcomes, software infrastructure, and AI services.
Key partners	Potential affiliate networks, retailers, VTO/body-measurement providers, cloud/AI vendors, pet brands.
Cost structure	Engineering, AI/API usage, cloud/database, analytics, and later customer acquisition.

13.1 Monetization experiments

- Ask interviewees which value they would pay for: more accurate recommendations, unlimited checks, visual try-on, household profiles, or return-history intelligence.
- Test price sensitivity with a simple range rather than committing to a price before interviews.
- Test whether users prefer a subscription or free tool supported by affiliate commerce.

- Do not build retailer/API monetization during the semester; use interviews to understand whether it could be credible later.

14. 15-Week Work Plan

Week	Milestone / deliverable
W1	Problem framing, market/competitor research, interview guide, research questions, success metrics.
W2	First 8-10 shopper interviews; capture current sizing workflow, returns, bracketing, and trust signals.
W3	Complete shopper interviews + survey; 3-5 exploratory pet-owner interviews; personas/customer journey.
W4	Synthesize findings; Value Proposition + Business Model Canvas v1; concept/pricing test; lock MVP scope.
W5	Figma end-to-end prototype; 8-10 usability tests; revise onboarding and recommendation explanation.
W6	Technical spike: product URL ingestion and normalization of size charts, fit notes, model data, material.
W7	Fit Passport schema + web foundation; profile, preferences, known-good items, product/outcome entities.
W8	Core MVP: onboarding, known-good closet, URL import, saved product and fit history.
W9	Baseline fit engine; ranked sizes, confidence, and grounded explanation.
W10	Post-purchase learning loop: keep/return/exchange and area-specific fit feedback.
W11	Regional sizing/product comparison; small pet-profile prototype; VTO proof of concept only if time.
W12	Integrated alpha; QA, privacy review, analytics, and beta protocol.
W13	Beta with ~10-15 target users; collect recommendation, confidence, usability, and repeated-use data.
W14	Analyze results; compare baselines; test effect of feedback; evaluate pet extension; BMC v2 and iteration.
W15	Final demo, findings, competitive positioning, roadmap, and required final written/reflection materials.

15. Success Metrics

Metric	How it is measured	What it tells us
Problem intensity	Interview frequency of size uncertainty, bracketing, returns, and purchase avoidance.	Is the pain frequent enough for a product?
Onboarding burden	Time and number of steps needed to create a usable passport.	Can users get value without measuring dozens of body points?
Recommendation agreement	Back-test recommendation vs known owned	Basic quality of the fit engine.

	size / fit outcome.	
Purchase confidence	1-5 rating before vs after seeing Fit Passport.	Does the product change the decision experience?
Explanation usefulness	User rating + qualitative feedback.	Does explainability build trust?
Feedback lift	Recommendation quality before vs after labeled fit outcomes.	Does the passport actually get more personal?
Repeat intent	Would user check another product / save the profile?	Early retention signal.
Pet portability signal	Pet-owner interviews + prototype reaction.	Does cross-domain expansion make sense?

16. Key Risks and Mitigations

Risk	Why it matters	Mitigation
Product-data availability	Retailer pages are inconsistent and may restrict automated scraping.	Use a limited supported set, public product information, manual fallback, and user confirmation of extracted data. The semester goal is feasibility, not internet-scale crawling.
Cold start	New users may not know body measurements.	Make measurements optional; start from 3-5 known-good garments and simple height/weight/fit-preference inputs.
Recommendation accuracy	A small course dataset cannot match enterprise fit platforms.	Benchmark only on a limited category, show confidence/assumptions, and frame the project as validation of the consumer-owned model rather than a claim of market-leading accuracy.
Scope creep	Apparel + shoes + rings + pets + VTO could become unfinishable.	Human tops/shirts/jackets are the required MVP. Pet is prototype only; all other verticals are roadmap.
Body/privacy sensitivity	Body measurements and purchase history can feel personal.	Collect the minimum needed data, make fields optional, explain why each field is used, and support deletion/export in the prototype design.
Visual try-on confusion	A realistic image can imply physical-fit accuracy that the model does not have.	Keep recommendation and visualization separate; label VTO as a visual approximation.
Pet variation	Breed and body proportions vary widely.	Use pet extension only to test workflow/architecture; do not claim general pet-sizing accuracy in the course.

17. Post-Semester Product Roadmap

Phase	Possible expansion
1 - Validate apparel	Improve recommendation model, support more apparel categories, test repeat use.
2 - Browser layer	Browser extension that shows Fit Passport directly on supported product pages.
3 - Household Fit Wallet	Multiple human and pet profiles under one account.
4 - Footwear	Foot length/width and known shoe-brand history; cross-brand size translation.
5 - Jewelry	Finger/ring measurements, regional conversion, known-good

	ring history; integrate rather than rebuild mature ring-sizing tech.
6 - Visual layer	Integrate existing VTO to visualize style while keeping physical size recommendation separate.
7 - Commerce/API	Affiliate links and APIs for retailers, marketplaces, and AI shopping agents.
8 - Measurement integrations	Optional mobile body scan or third-party measurement providers such as 3D body scanning services.

18. What We Aim to Learn

The most important outcome of the semester is not a polished fashion app. It is evidence about whether consumers want to own a reusable fit identity and whether a lightweight personal history can improve the way they make size decisions across stores. If interviews show that users will not maintain the profile, or back-testing shows that known-good items do not add enough value, that is a meaningful result. If the profile improves confidence and fit decisions, the project has a clear path to a larger product.

The vision is broad - one fit identity for people, pets, and eventually multiple product categories - but the implementation is intentionally narrow. A focused apparel web MVP gives the team a realistic 15-week deliverable while still testing the core startup thesis: fit data should accumulate around the shopper, not disappear into separate retailer experiences.

References

- [1] National Retail Federation. "2025 Retail Returns Landscape." Oct. 15, 2025. Key figures: \$849.9B projected total U.S. retail returns and 19.3% of online sales returned.
- [2] Shin, E., & Baytar, F. "Apparel Fit and Size Concerns and Intentions to Use Virtual Try-On." Clothing and Textiles Research Journal, 2014. DOI: 10.1177/0887302X13515072.
- [3] Hernández, N., Mattila, H., & Berglin, L. "Can Virtually Trying on Apparel Help in Selecting the Correct Size?" Clothing and Textiles Research Journal, 2019. DOI: 10.1177/0887302X19856117.
- [4] "Do Fit Opinions Matter? The Impact of Fit Context on Online Product Returns." Information Systems Research, 2020. DOI: 10.1287/isre.2020.0965.
- [5] Grand View Research. "Virtual Fitting Room Market Size Report, 2025-2030." Market estimate: \$8.5B in 2026; \$20.6B forecast in 2030.
- [6] True Fit. "About" and "How It Works." Current product pages accessed Aug. 7, 2026. Describes fit intelligence using purchase/return outcomes and a global product/brand dataset.
- [7] Bold Metrics. "Fit and Sizing Technology" and "Smart Size Chart." Accessed Aug. 7, 2026. Describes 50+ body measurements, 250M+ digital twins, and 750M+ fit simulations.
- [8] 3DLOOK. "Our Technology." Accessed Aug. 7, 2026. Describes body measurement capture using two smartphone photos.
- [9] Sizebay. "Virtual Try-On," "Size Guide," and current Terms & Conditions. Accessed Aug. 7, 2026. Describes retailer-integrated size recommendation, fit preference, regional size charts, and footwear support.
- [10] Google Labs. "Try on looks and discover your style with Doppl." Jun. 26, 2025. Describes digital-self virtual try-on and AI-generated outfit videos; notes that fit and appearance may not always be accurate.
- [11] Perfect Corp. "Virtual Ring Try On & Online Sizer" and ring sizing guide. Accessed Aug. 7, 2026. Describes web/mobile ring measurement, regional conversion, and AR try-on.
- [12] American Pet Products Association (APPA). "U.S. Pet Industry Reaches \$158 Billion in 2025, Poised for Continued Growth in 2026." Mar. 26, 2026. Reports 95M U.S. pet-owning households, 71M dog-owning, and 53M cat-owning households.
- [13] Grand View Research. "Pet Clothing Market Size, Growth Report, 2026-2033." Market estimate: approximately \$2.3B in 2026.
- [14] Ruffwear. "The Size is Right: How to Measure for Your Dog's Gear." May 21, 2024. Describes girth-based sizing for harnesses, apparel, and packs.
- [15] Chen, S. X. et al. "SiCo: A Size-Controllable Virtual Try-On Approach for Informed Decision-Making." arXiv:2408.02803, 2024.
- [16] Ning, Y. et al. "FitVTON: Fit-aware Virtual Try-On via Body-Garment Size Control." arXiv:2606.12012, 2026.